

Differentiated Carbon Prices or International Transfers?

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“Efficiency is decisive; distribution can be sorted out separately.”

Differentiated prices: each country sets its own price according to income and development needs.

“Large international transfers will never happen; differentiation is the second best.”

This paper: the disagreement is (mostly) a false one

Equivalence. For any schedule of differentiated prices (p_i), there is a uniform price p^* plus an allocation of emission rights (r_i) leaving every country exactly as well off.

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⇒ “Prices vs. rights” is **not a debate about instruments** but about **how to split the gains from one**.

Rationales for differentiated prices

Why differentiate? Three arguments, only one of which we address

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But **an allocation of emission rights is a transfer**, denominated in tonnes rather than dollars, cleared by the very market that sets the uniform price. **Infeasibility would have to bite on rights too** — it is no argument against trading per se.

Theory

The static model

Countries i , one period, no uncertainty. Facing carbon price p_i , country i abates at cost a_i :

$$c_i = y_i + t_i - a_i, \quad t_i = \text{international transfer (+ if received)}.$$

Welfare u_i depends on c_i and on global emissions $E = \sum_j e_j n_j$, **fixed at the agreed target E^*** .

The question is *not* “what is the optimal target?” but “**given the target, who pays, and how efficiently?**”

Temperature is *identical* under both regimes; only distribution and efficiency differ.

p^* : the uniform price attaining E^* , i.e. $E^* = \sum_j e_j(p^*) n_j$. Shorthands $e_i^* := e_i(p^*)$, $a_i^* := a_i(e_i^*)$.

A **situation** is a pair $s = (t_i, p_i)$. Two anchors:

Benchmark $s^* = (0, p^*)$: uniform price, no transfer.

Autarky $s^a = (0, p_i)$: differentiated price, no transfer.

Step 1: the equivalent variation of autarky

$V_i(p_i)$: the lump sum which, paid *at the benchmark price*, makes i as well off as in autarky:

$$u_i(V_i(p_i), p^*) = u_i(0, p_i) \implies V_i(p_i) = c_i^a - c_i^* = a_i^* - a_i.$$

First-order Taylor expansion around e_i^* , where the optimising country sets $\frac{da_i}{de}(e_i^*) = -p^*$:

$$V_i(p_i) \approx (e_i - e_i^*) p^* = (p^* - p_i) e_i^* \varepsilon^* \quad \varepsilon^* = -\frac{de_i/e_i^*}{dp/p^*} : \text{price elasticity at } p^*.$$

$$g_i^s = \underbrace{t_i}_{\text{cash received}} + \underbrace{V_i(p_i)}_{\text{gain from facing } p_i \neq p^*}$$

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Sign is intuitive: if $p_i < p^*$, i abates less in autarky $\Rightarrow$ moving it to p^* requires **compensation** $V_i > 0$.

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Equivalent transfer of an arbitrary situation $s = (t_i, p_i)$, relative to the benchmark:

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One money-metric number at a common price $\Rightarrow$ comparable across regimes differing in *both* t and p .

► Why not an equivalent price?

Step 2: the correspondence between prices and rights

Cap-and-trade at p^* , each person in i endowed with rights r_i . Net monetary transfer to i :

$$t_i^* = (r_i - e_i^*) p^* \quad (i \text{ sells if } r_i > e_i^*, \text{ buys otherwise}) \quad \implies \quad g_i^* = t_i^*.$$

Autarky, from Step 1: $g_i^a = V_i(p_i) \approx (p^* - p_i) e_i^* \varepsilon^*$. Setting $g_i^* = g_i^a$:

$$(r_i - e_i^*) p^* \approx (p^* - p_i) e_i^* \varepsilon^*$$

$$\iff r_i \approx \left(1 + \left(1 - \frac{p_i}{p^*}\right) \varepsilon^*\right) e_i^* \iff p_i \approx \left(1 + \frac{e_i^* - r_i}{e_i^* \varepsilon^*}\right) p^*.$$

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The key formula for policy design: given an autarky price p_i , how many rights leave welfare unchanged?

Low autarky price ($p_i < p^*$) $\Rightarrow$ **generous allocation** $r_i > e_i^*$ compensates the extra abatement burden.

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$\Rightarrow$ Any differentiated-price arrangement **maps onto** a cap-and-trade allocation, and back.

Natural follow-up: if the two can always be made equivalent, *is there any reason to prefer one?* Yes.

Step 3: the uniform price weakly dominates, for *every* country

Give country i rights equal to its *autarky* emissions, $r_i = e_i$. Net transfer $\tilde{t}_i = (e_i - e_i^*)p^*$.

Convexity of abatement costs:

$$a_i - a_i^* \geq (e_i - e_i^*) \frac{da_i^*}{de} = -(e_i - e_i^*)p^* \quad \implies \quad \tilde{t}_i \geq a_i^* - a_i = V_i(p_i) = g_i^a.$$

Hence consumption under the cap-and-trade:

$$\tilde{c}_i = y_i + \tilde{t}_i - a_i^* \geq y_i - a_i = c_i^a$$

for **every** country **simultaneously**.

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This is Diamond & Mirrlees (1971) production efficiency applied to climate: absent other distortions, direct transfers beat differentiated factor prices as a redistributive instrument.

The dynamic case: the equivalence breaks

Add time t ; each variable becomes a trajectory. β_t = discount factor, $R_i = \sum_t r_{it}$ = carbon budget.

Welfare equivalence now requires equality of *entire discounted streams*, $G_i^U = G_i^A$:

$$\sum_t (r_{it} - e_{it}^*) \beta_t p_t^* \approx \sum_t (p_t^* - p_{it}) e_{it}^* \varepsilon_t^* \beta_t.$$

In the static case p^* was a *common factor* and cancelled. **Here it does not**: p_t^* sits inside both sums.

A generous allocation in a **low-price** period is worth **less** than the same allocation when prices are high.

$\Rightarrow$ **No direct correspondence between the intertemporal carbon budget R_i and welfare**. Timing matters, and timing depends on the full trajectory $(p_t^*)_t$.

$$\rho_i \mathcal{B} - \sum_t e_{it}^* p_t^* \beta_t \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* p_t^* \beta_t, \quad \mathcal{B} = \sum_t r_t \beta_t p_t^*.$$

Analogous to the static formula, but $\mathcal{B}$ still requires the full price path $\Rightarrow$ **not tractable without a model**.

The dynamic case: the equivalence breaks

Add time t ; each variable becomes a trajectory. β_t = discount factor, $R_i = \sum_t r_{it}$ = carbon budget.

Welfare equivalence now requires equality of *entire discounted streams*, $G_i^U = G_i^A$:

$$\sum_t (r_{it} - e_{it}^*) \beta_t p_t^* \approx \sum_t (p_t^* - p_{it}) e_{it}^* \varepsilon_t^* \beta_t.$$

In the static case p^* was a *common factor* and cancelled. **Here it does not**: p_t^* sits inside both sums.

A generous allocation in a **low-price** period is worth **less** than the same allocation when prices are high.

⇒ **No direct correspondence between the intertemporal carbon budget R_i and welfare**. Timing matters, and timing depends on the full trajectory $(p_t^*)_t$.

Special case 1 — homothetic prices, fixed budget shares ($p_{it} = \pi_i p_t^*$ and $r_{it} = \rho_i r_t$):

$$\rho_i \mathcal{B} - \sum_t e_{it}^* p_t^* \beta_t \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* p_t^* \beta_t, \quad \mathcal{B} = \sum_t r_t \beta_t p_t^*.$$

Analogous to the static formula, but $\mathcal{B}$ still requires the full price path ⇒ **not tractable without a model**.

Special case 2: Hotelling — and why it is the right benchmark

Fixed aggregate carbon budget + perfect foresight + efficient capital markets $\Rightarrow$ the carbon price rises at the rate of return on capital. Discounting at that same rate, $\beta_t p_t^* = p_0$ is constant.

p_t^* then factors out of both sums, and the dynamic condition collapses to:

$$R_i - E_i^* \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t \quad E_i^* = \sum_t e_{it}^*.$$

Structurally identical to the static formula: the intertemporal carbon budget R_i plays the role of the single-period allocation, and the homothetic price factor π_i the role of p_i/p^* .

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$\Rightarrow$ If the approximation fails here, it fails everywhere. Hence we take it to a calibrated model: not “does it hold in theory?” but “how far do simulated welfare effects deviate, and what do the deviations reveal?”

Numerical test

Taking the formula to a calibrated model

The formula rests on approximations a calibrated model need not satisfy: constant elasticities, homothetic price paths, perfect capital markets, a first-order Taylor expansion. **Whether it predicts well is an empirical question.**

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Grid search over (π_i, ρ_i) for eight representative entities: **USA, Russia, EU27, China, Turkey, India, Nigeria, DR Congo**.

The two scenarios

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Extensions: abatement rates $\mu_i > 1$ allowed (**CDR**), MAC extrapolated beyond 100%; **viability filter** flagging RoW fossil subsidies ($\omega_i \pi_i > 1$) and negative RoW rights ($\rho_i > 1 / pop_i$). ► Detail

Reading the heatmaps

One heatmap per country over (π_i, ρ_i) :

x: autarky price factor π_i .

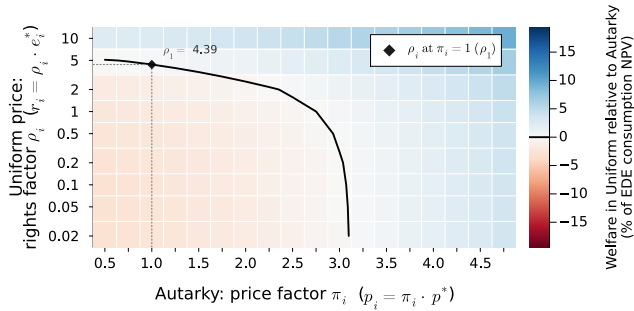
y: rights multiplier ρ_i (log scale).

colour: welfare gain of *uniform* over *autarky*, in % of the NPV of EDE consumption.

Blue = uniform price with transfers dominates; **red** = autarky dominates.

The **empirical indifference curve** separates the two: for each π_i , the **minimum allocation** at which i weakly prefers the uniform regime.

Its intercept at $\pi_i = 1$ is the **non-losing allocation** ρ_1 , which the static formula predicts.



Non-losing rights allocation: prediction vs. simulation

Predicted ρ_1 = emission share / population share (NPV) = **per-capita emissions relative to the world average**. [► All heatmaps](#)

Country	Emission share 2030 (%)	Population share 2030 (%)	Emissions p.c. 2030 (tCO ₂)	Predicted ρ_1	Actual ρ_1	Deviation
USA	15.9	4.1	14.12	4.44	4.39	−1.1%
RUS	3.8	1.7	8.26	2.89	2.85	−1.4%
CHN	29.9	17.3	6.34	0.95	0.67	−29.5%
TUR	1.2	1.1	4.18	1.65	1.83	+10.9%
EU27	5.6	5.2	3.97	0.30	0.23	−23.3%
IND	8.3	17.8	1.72	0.87	0.91	+4.6%
NGA	0.4	3.1	0.46	0.18	0.15	−16.7%
COD	0.0	1.4	0.05	0.03	0.02	(−33%)

ρ_1 spans more than two orders of magnitude: 0.02 (DR Congo) to 4.39 (USA). In all eight cases there exists an allocation above which the uniform regime weakly dominates autarky.

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The formula predicts that threshold well for six of eight. Only China and EU27 need a separate story.

Where the static formula holds

USA (−1.1%) and Russia (−1.4%): near-exact — both high-income, high per-capita emitters.

Uniform-price scenario, DR Congo ($\rho = 0.5$)								
Variable	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
r_i (GtCO ₂)	0.221	0.185	0.131	0.099	0.043	0.0	0.0	0.0
e_i (GtCO ₂)	0.005	0.007	0.008	0.008	0.004	0.0	0.0	0.0
Transfer (bn USD)	9.2	20.1	25.4	25.9	15.4	0.0	0.0	0.0

\$9bn in 2030 $\approx$ 13% of current GDP, against actual emissions of 0.005 GtCO₂. No autarky price comes close.

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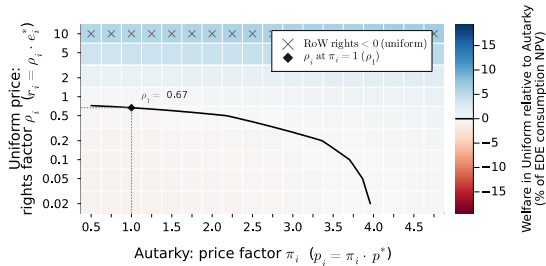
Nigeria and the DRC best illustrate **Diamond–Mirrlees**. Negligible per-capita emissions and almost no abatement capacity $\Rightarrow$ the **autarky price is nearly irrelevant**. **What matters is the allocation**.

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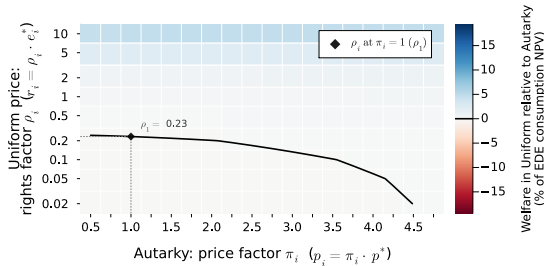
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China and the EU27: two mechanisms the static model abstracts from

China: $\rho_1 = 0.67$, predicted 0.95 (-29.5%)



EU27: $\rho_1 = 0.23$, predicted 0.30 (-23.3%)



Free-riding does not pay at this scale. With $\sim 30\%$ of 2030 emissions, a low Chinese price forces a large compensating adjustment on the RoW to hold the cap.

That feeds back through **trade and capital markets**, offsetting part of the domestic gain from $\pi_i < 1$.

$\Rightarrow$ China accepts a **below-proportional** allocation to access the global market.

Cheap abatement already spent through the ETS and domestic regulation.

In autarky, meeting the 1.8°C budget puts the EU on the **expensive part of its MAC curve at every price level**.

$\Rightarrow$ Access to cheaper reductions abroad is valuable *even at a reduced allocation*: autarky is costly across nearly the whole range of π_i .

Application

Equivalent rights for two live proposals

[Wolfram et al. \(2025\)](#): voluntary coalition on a *tiered* price floor (higher for richer members) + shared CBAM; ~\$200bn/yr of domestic carbon revenue.

[Duflo et al. \(2025\)](#): “grand bargain” — eligibility for climate-damage compensation conditional on a graduated domestic price (\$10 LICs / \$30 lower-MICs / \$50 upper-MICs).

Fed into NICE as country-level π_i ; we solve for the ρ_i making each country **indifferent** to trading its proposal price for uniform pricing.

Country	Wolfram			Duflo		
	ΔT	Welfare	ρ_{Wolfram}	ΔT	Welfare	ρ_{Duflo}
CHN	+7.4	529.46	1.306	+7.4	530.02	1.514
COD				−32.6	165.56	<0.0001
EU27	+32.4	1247.15	<0.0001	−42.6	1247.5	0.043
IND	−17.6	255.88	1.266	−12.6	254.58	0.836
NGA				−12.6	265.86	<0.0001
RUS				−42.6	857.24	8.043
TUR				+7.4	760.05	0.574
USA				−42.6	1171.3	11.04
World rights reallocation			50.2%			48.3%
World temp. 2100, autarky → uniform			2.14 → 1.757°C			1.984 → 1.757°C
World welfare gain			0.26%			3.2%

ΔT : gap between domestic autarky tax and global uniform price in 2030. “<”: uniform welfare already beats autarky over the tested ρ range.

Conclusion

What the result does *not* establish

A first-best statement. The dominance result assumes away every other distortion. Where the second-best arguments bite — market power, pre-existing sectoral taxes, divergent discount rates (Anthoff et al., 2022; Babiker et al., 2004) — differentiation can be optimal in its own right, and we are silent. ▶ The three arguments

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One deviation at a time. Each heatmap asks “what if *i alone* goes it alone?”. The politically relevant case — **several large emitters differentiating simultaneously** — is outside the grid.

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A first-best statement. The dominance result assumes away every other distortion. Where the second-best arguments bite — market power, pre-existing sectoral taxes, divergent discount rates (Anthoff et al., 2022; Babiker et al., 2004) — differentiation can be optimal in its own right, and we are **silent**. ▶ The three arguments

Negishi recycling is imposed, not estimated. Domestic revenue returns in inverse proportion to marginal utility, neutralising the within-country distributive effect *by construction*. That is what isolates the pure efficiency channel — but it means our numbers say **nothing about domestic incidence**, which is itself the contested political choice.

The Rest of the World is a single aggregate, grandfathered on baseline emissions, absorbing all adjustment through one price p_{-i} . **Coalition formation, leakage and terms-of-trade effects among third countries are invisible** to us.

One deviation at a time. Each heatmap asks “what if *i alone* goes it alone?”. The politically relevant case — **several large emitters differentiating simultaneously** — is outside the grid.

The dynamic equivalence leans on Hotelling. Imperfect foresight or frictional capital markets break the constancy of $\beta_t p_t^*$ and **reopen the timing problem** the static formula cannot see. ▶ Why Hotelling

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Any schedule of differentiated carbon prices has a welfare-equivalent uniform price plus rights allocation — and that allocation is weakly better for every country at once (Diamond–Mirrlees).

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► Limitations

Thank you for your attention!

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Appendix

Viability filter and the RoW price [► Go back](#)

Two boundary conditions evaluated in *every* year of 2030–2100:

1. **Market-distorting subsidies** ($\omega_i \times \pi_i > 1$): the target country claims more than 100% of the global price burden. To compensate, the model is forced to push $p_{-i} < 0$ — economically, a **fossil-fuel subsidy for the RoW**.
2. **Negative emission rights** ($\rho_i > 1 / pop_i$): granting i its multiplier uses more than the whole global budget, forcing the RoW into **negative rights**. Binds for large populations: countries above 10% of world population (China, India) trigger it for any $\rho_i > 10$.

RoW price identity, the starting point of the root-finding:

$$p^* = \omega_i p_i + (1 - \omega_i) p_{-i}, \quad \omega_i = e_i / E^* \quad \implies \quad p_{-i} = p^* (E^* - e_i \pi_i) / e_{-i}.$$

This linear approximation assumes emission-weighted price averaging delivers the same global abatement — false with non-linear MACs, hence the **Newton–Raphson** correction to $< 0.1\%$ relative error on the cap.

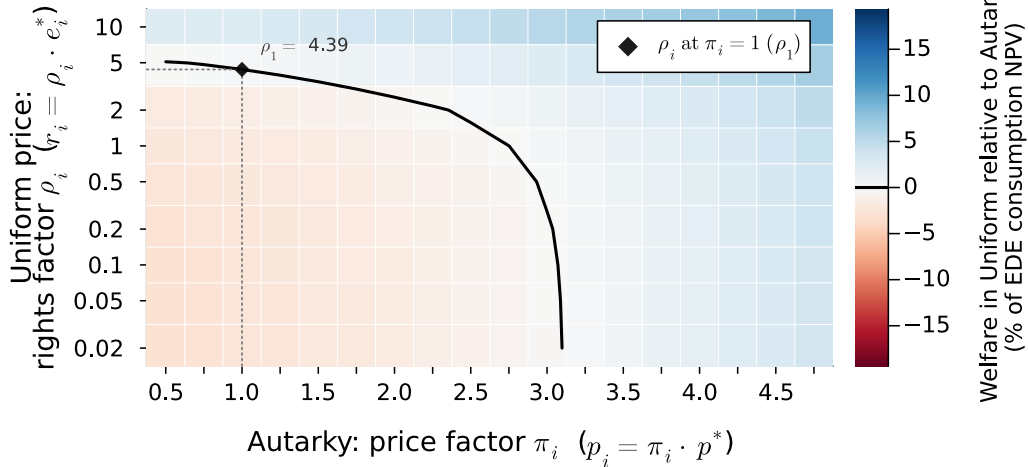
Autarky scenario, United States ($\pi = 2$) [Go back](#)

Variable	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
p_i (USD/tCO ₂)	85.2	225.8	411.7	571.4	803.3	961.9	952.3	942.8
E_i (GtCO ₂)	4.31	2.58	0.63	-0.53	-1.75	-2.22	-1.86	-1.52

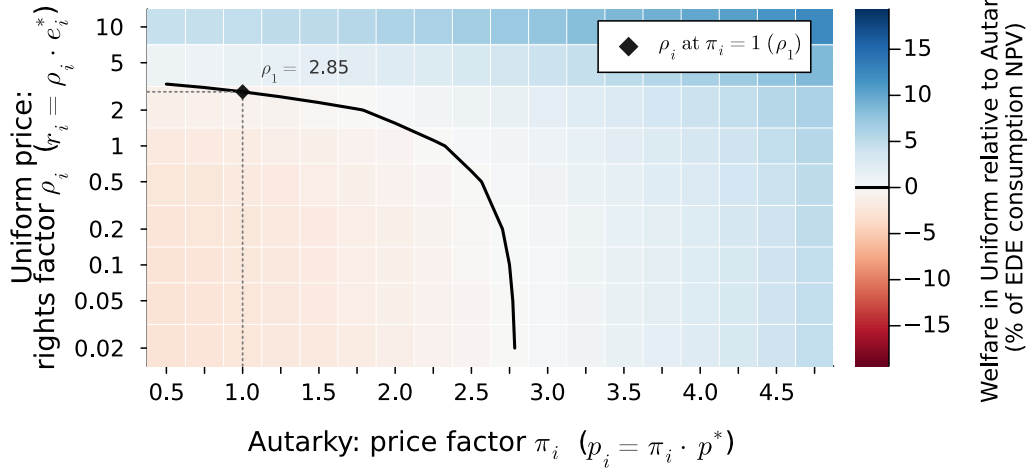
Net-negative US emissions from the 2060s: abatement rates $\mu_i > 1$, i.e. [CDR deployment](#), with the MAC function extrapolated beyond the 100% boundary.

Standard IAMs constrain $\mu_i \in [0, 1]$; enforcing that bound crashes the grid search under extreme policy parameters.

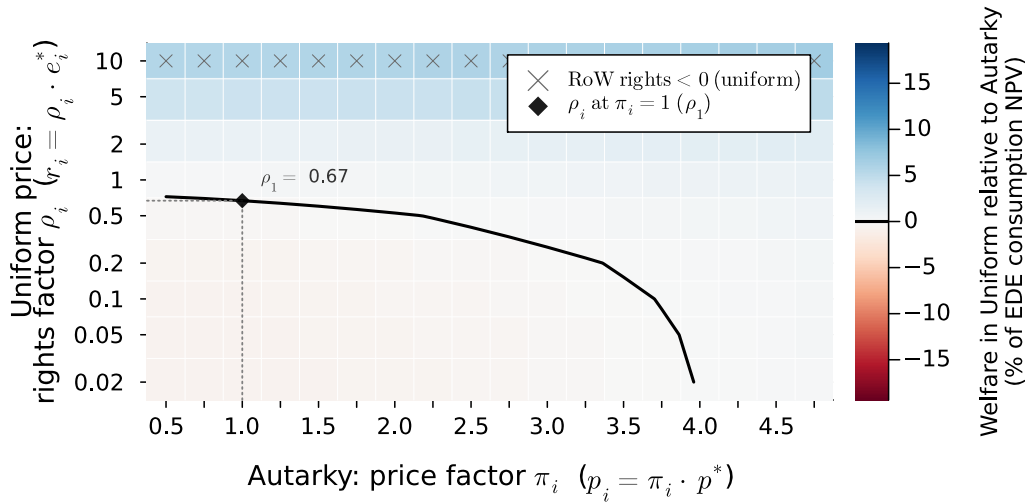
United States



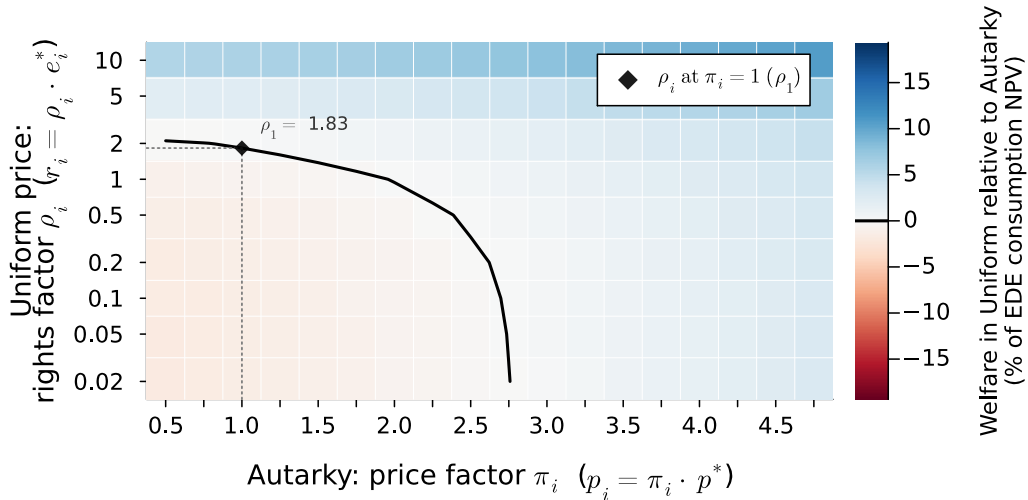
Russia



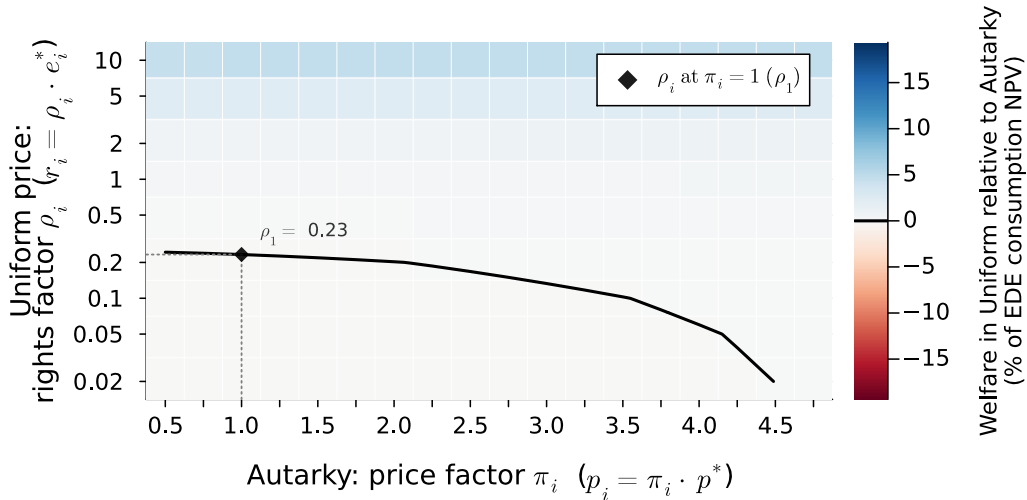
China



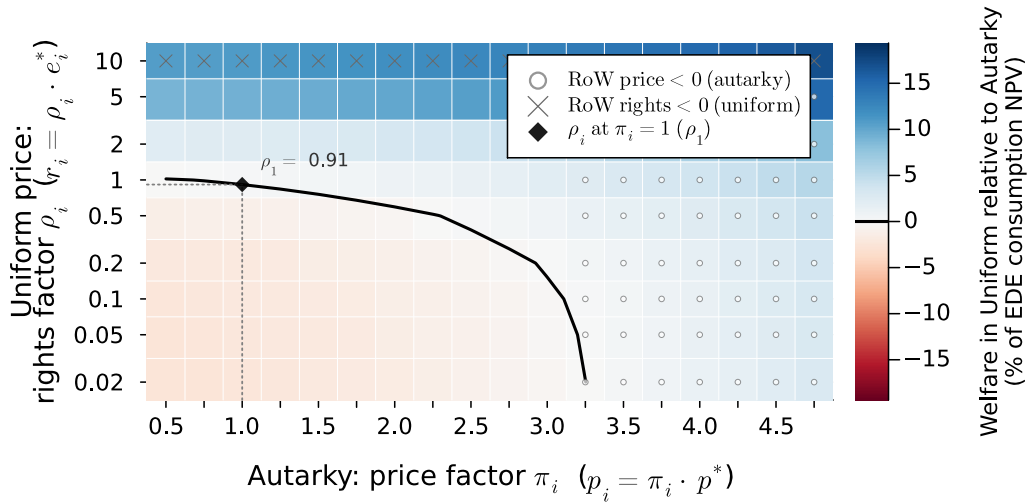
Turkey



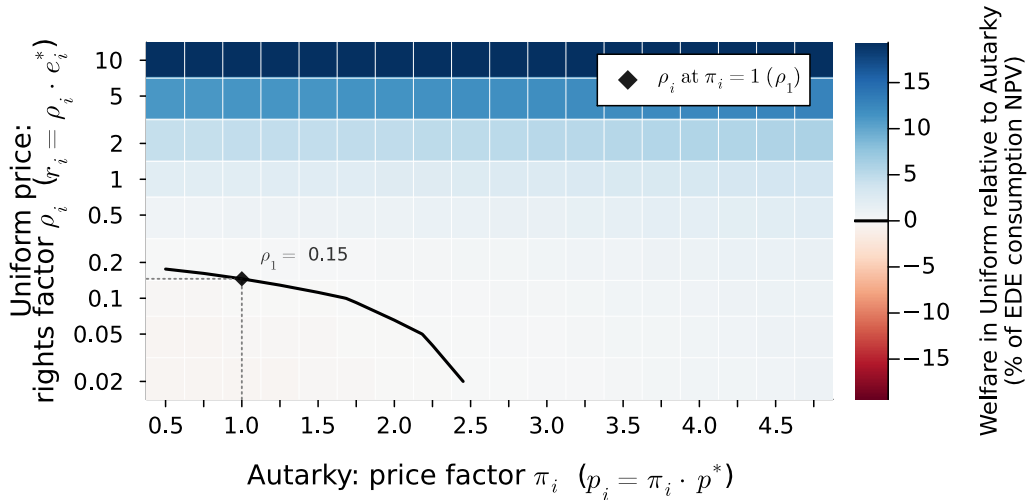
European Union (EU27)



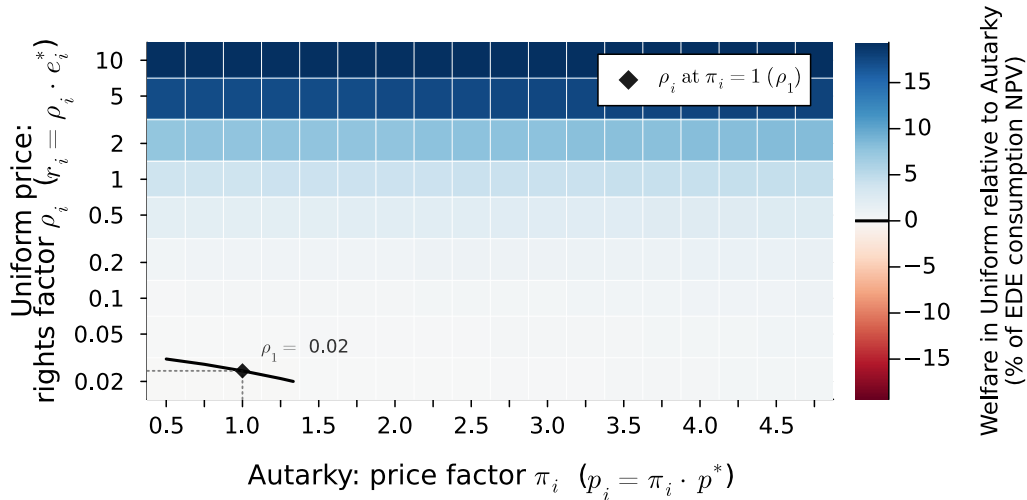
India



Nigeria



Democratic Republic of Congo



The equivalent price, and why we do not use it [▶ Go back](#)

One could instead define an **equivalent price** p_i^* such that $(0, p_i^*)$ is welfare-equivalent to (t_i, p_i) :

$$t_i + (p^* - p_i)e_i^*\varepsilon^* \approx (p^* - p_i^*)e_i^*\varepsilon^* \quad \implies \quad p_i^* \approx p_i - \frac{t_i}{e_i^*\varepsilon^*}.$$

Problem: p_i^* can be *negative* when the transfer is large, which has no natural interpretation as a carbon price.

The equivalent transfer g_i^s does not share this limitation $\implies$ we use it as the primary welfare metric.

Last slide (needed)